

Chapter 4

Compositional and Chemical Control of Volatile Polymer-Electrolyte Memristive Dynamics

4.1 Introduction and Scope

?? established the memristive response of the *Super Yellow/Hybrane*/LiOTf formulation and 2-T architecture across assay-specific junctions. This chapter measures how changes in composition, electrolyte chemistry, electrode, and drive protocol alter the volatile dynamics of the same three-component architecture in a broader device corpus.

Three families of measurement recur on the same 2-T geometry throughout this chapter and the next: (i) current–voltage hysteresis, which reports the steady-state switching window; (ii) potentiation as a function of the number of applied write pulses, which reports how the conductance is built up; and (iii) depotentiation as a function of the delay before

read-out, which reports how a written state fades. Introduced as a named set in ??, they are the shared, abundant basis of the comparative work. The richer proof-of-concept synaptic protocols of ?? (excitatory post-synaptic current, separated short- and long-term memory, and spike-timing dependent plasticity) are *not* repeated here; where invoked, they serve only as formulation-level prior evidence.

Scope of the comparisons. The *Super Yellow*/PEO/LiOTf composition grid contains typically two to four nominally identical devices per cell and supports quantitative comparisons. After data-quality screening, the host, anion, and cation contrasts contain at most one or two devices per matched cell and are used only as an illustrative survey. The electrode comparison is likewise illustrative and is available only at the lead composition (section 4.8). Drive protocol is analysed separately because it changes the apparent fading-memory timescale and therefore limits comparisons between protocols.

4.2 Materials, Devices, and Measurement Protocol

The devices retain the architecture of ??: a vertical two-terminal sandwich of ITO (bottom, transparent), a solution-processed composite active layer, and a thermally evaporated silver top electrode, with a junction area of $8.25 \times 10^{-2} \text{ cm}^2$. *Super Yellow* is retained as the semiconducting component in every device of this chapter, so that the electronic-transport role is held fixed and the variation is confined to the ion-transport phase.

Two design axes are varied. The *composition* axis varies the mass fraction of the ion-transport polymer poly(ethylene oxide) (PEO) and of the lithium triflate salt (LiOTf) relative to the semiconductor. The *chem-*

istry axis substitutes the ion-transport host, the salt anion (triflate or bis(trifluoromethanesulfonyl)imide, TFSI), and the alkali cation (Li^+ , Na^+ , K^+ , as the triflate or TFSI salt). The comparative work replaces the hyperbranched polyester-amide *Hybrane* host of the proof-of-concept device (??) with two hosts used here at the same level: linear PEO and the hyperbranched polyether trimethylolpropane ethoxylate (TMPE), so that the host axis contrasts a linear against a branched polyether at matched chemistry. Every device reported *quantitatively* uses the silver electrode. A separate gold-electrode corpus at the lead composition provides a third, illustrative axis (section 4.8). It compares an active with an inert electrode, measures drive-polarity dependence on gold, and provides an independent-electrode check on the chemistry comparisons.

Control for film thickness. Adding ion-transport polymer thickens the solution-processed film, so most higher-PEO cells were spun faster to reduce thickness differences across the grid (1500 rpm–3500 rpm). This compensation was incomplete and was not applied uniformly to every comparison batch. In the silver composition series ($n = 30$), median thickness rises from ≈ 227 nm at PEO 0.3 to ≈ 298 nm at PEO 1.2 (Pearson $r \approx 0.68$; fig. 4.1). Several controls separate this covariation from the composition result. The reported responses are dimensionless ratios or timescales; activation voltage remains within 2.2 V–2.4 V across the full $2.6\times$ thickness range ($r \approx -0.06$); and a 38% within-cell thickness spread leaves the fading-memory time essentially unchanged. The partial correlation between thickness and $\log t_{1/2}$, controlling for PEO, is $\approx +0.05$, whereas the association with composition, controlling for thickness, is ≈ -0.42 . Thus, thickness covaries with polymer loading, while the dynamics retain an association with composition after thickness control. Full statistics and the audit of other fabrication variables are given in Appendix C.

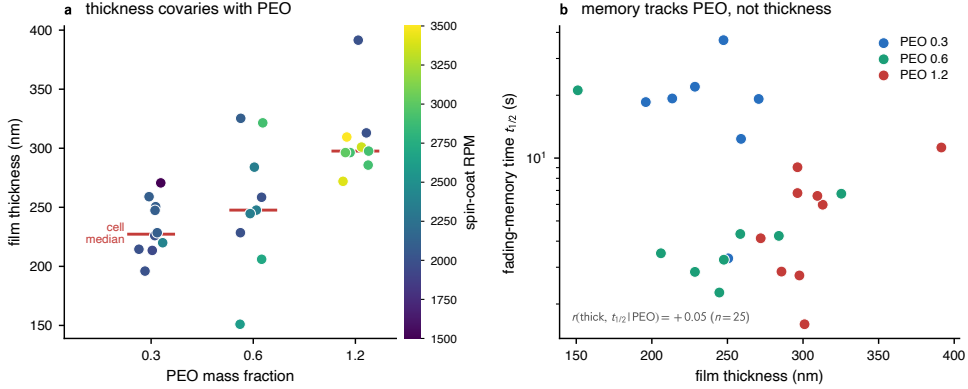


Figure 4.1: Film thickness and fading-memory time in the silver composition series ($n = 30$). (a) Film thickness rises with PEO fraction (red bars are cell medians); higher-PEO cells were spun faster (colour) to reduce this difference, although residual covariation remains. (b) Within each PEO band, $t_{1/2}$ is approximately flat against thickness; the partial correlation between thickness and $t_{1/2}$, controlling for PEO, is ≈ 0 .

Drive and read protocols. Because the comparisons below turn on them, the electrical protocols are stated explicitly, and they are *not* identical across the three measurements (table 4.1). Hysteresis uses a triangular voltage sweep. The potentiation (N -pulse build-up) was acquired with a 3 V write and 1.5 V read at a variable pulse count ($N \leq 1000$); the depotentiation (fading-memory decay) used a fixed train of 30 write pulses at a 4 V write and 2 V read. The inter-pulse interval is effectively common (0.104 s for potentiation, 0.103 s for depotentiation), but the *write amplitude differs* between the two (3 V vs 4 V), a point that matters when the two are later composed (section 4.10, ??). The read voltage is above the ionic activation threshold established in ??, a point returned to in section 4.6. An older 2022 device generation used a 6 V write with a 3 V read; the modern and old amplitudes are never pooled, and the same-device contrast between two amplitudes is itself the drive-protocol result (section 4.6).

Table 4.1: The three common measurements use distinct drive protocols (modern *Super Yellow*/PEO/LiOTf/Ag corpus). Potentiation and depotentiation differ in write amplitude, so the chapter never pools them and ?? composes them only as an explicit assumption.

Measurement	Write	Read	Pulse train
Hysteresis (HYST)	triangular sweep	—	—
Potentiation (PULSES)	3 V	1.5 V	variable $N \leq 1000$, 0.104 s
Depotentiation (DELAYTIME)	4 V	2 V	30 pulses, 0.103 s
Protocol overlay (v114)	3 V, 6 V	1.5 V, 3 V	30 pulses

4.3 Methods of Analysis

Hysteresis curves are summarised by the on–off conductance ratio and by a normalised loop area; potentiation curves by the per-decade growth of the conductance ratio with pulse number, by the peak ratio, and by whether the response turns over (decreases) at the highest pulse counts; and depotentiation curves by a fading-memory time constant.

For the depotentiation (fading-memory) decay, the comparative archive is refitted with the more general Kohlrausch (stretched-exponential) function

$$\Delta G(t_{\text{wait}}) = \Delta G_0 \exp \left[- \left(\frac{t_{\text{wait}}}{\tau} \right)^\beta \right] + \Delta G_\infty, \quad (4.1)$$

with characteristic relaxation time τ , stretch exponent $0 < \beta \leq 1$, initial change ΔG_0 , and residual ΔG_∞ [1]. Two practical points follow from the data. First, the eight-to-thirteen-point decay curves often under-constrain β and τ jointly; the fitted τ is therefore reported only where the fit is well identified, and a robust, model-free half-enhancement time $t_{1/2}$ (the delay at which the conductance enhancement has fallen to half its initial value) is used as the primary timescale. Where both are available they agree. Both

are *apparent* timescales determined by the material and by the amplitude used to write and read the state. Section 4.6 establishes this protocol dependence; the composition series and chemistry comparisons therefore each use a fixed protocol. Second, fitted β values lie predominantly in the range 0.6–0.9, consistent with stretched relaxation and the heterogeneous distribution of activation barriers invoked in ??.

Data-quality control and aggregation. Two screening layers are applied before any number is reported: erratic or non-physical measurements are flagged and excluded, and a per-device visual review stores per-curve verdicts (used as measured, cleaned, or discarded) in a curation registry applied automatically by the analysis scripts. All timescales are recomputed after this screening, because the instrument-side fit is not robust to anomalous points. Summaries then follow a conservative curve→pixel→device→cell ladder of medians, and the representative curves of fig. 4.2 are drawn by script from devices near their cell medians, not hand-picked. The full raw-to-summary trail, the aggregation ladder, and the stretched-exponential identification thresholds are documented in Appendix C.

4.4 Composition Control of the Dynamics

The quantitative core of this chapter is the *Super Yellow*/PEO/LiOTf composition grid, in which the PEO mass fraction is varied over 0.3, 0.6 and 1.2 and the salt mass fraction over 0.045, 0.09 and 0.18, with two to four nominally identical silver-electrode devices per cell. Each measurement was acquired under its own fixed protocol (table 4.1); the composition contrasts below are therefore always drawn *within* a single measurement, never by pooling the 3 V potentiation with the 4 V depotentiation. Before the grid is summarised, fig. 4.2 shows what the three

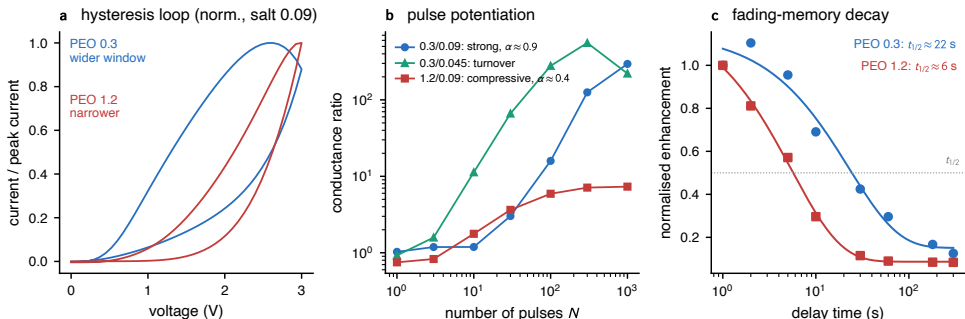


Figure 4.2: Representative raw curves of the three common measurements, on devices chosen near their cell medians (*Super Yellow*/PEO/LiOTf/Ag). (a) Hysteresis loops, each normalised to its own peak current so the comparison is the loop *openness* (the switching window), not absolute conductance: the low-PEO device (PEO 0.3, blue) opens a wider window than the high-PEO one (PEO 1.2, red), which carries more current but switches over a narrower range. (b) Pulse potentiation (conductance ratio vs pulse number, log-log): a strong near-linear accumulation ($\alpha \approx 0.9$), a low-salt cell that *turns over* and then declines, and a weak, compressive high-PEO response ($\alpha \approx 0.4$). (c) Fading-memory decay (enhancement vs delay, normalised to the first read) with stretched-exponential fits and the half-enhancement level dotted: the low-PEO device crosses $t_{1/2}$ far later (≈ 22 s) than the high-PEO one (≈ 6 s).

underlying measurements actually look like on representative devices, so that the on-off ratio, the growth exponent, the turnover, and the fading-memory time can be read off real curves before they are reduced to per-cell numbers. Three trends, mutually consistent and each visible across the grid, define the result.

First, the steady-state switching window contracts as the PEO fraction rises. At fixed salt mass fraction 0.09, the median on-off ratio falls from ≈ 3.3 at PEO 0.3 to ≈ 2.4 at PEO 1.2, and the normalised loop area falls correspondingly from ≈ 0.33 to ≈ 0.18 ; the activation voltage is, by contrast, essentially flat across the whole grid (2.2 V–2.4 V), indicating that composition tunes how *much* the conductance moves, not the voltage at which it begins to move. The widest window in the grid sits in the

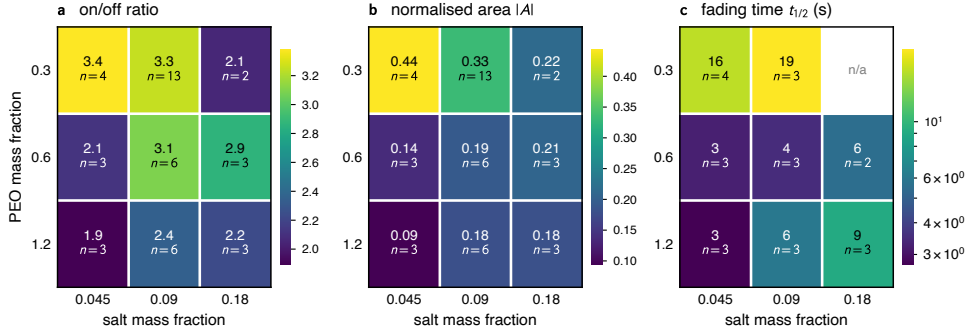


Figure 4.3: Composition control of the steady-state and dynamical response of *Super Yellow*/PEO/LiOTf devices. The switching window (a, b) and the fading-memory time (c) both decrease as the PEO mass fraction increases at fixed salt content.

low-PEO row, with the largest loop area at the low-salt corner (PEO 0.3, salt 0.045: area ≈ 0.44); the per-sweep on–off ratio is strongly right-skewed there (cell median ≈ 3.4 but individual sweeps reaching ~ 40), so robust cell medians, not means, are reported throughout to avoid letting a few outlying sweeps define a cell.

Second, potentiation weakens, and changes shape, as the composition is varied (fig. 4.5). Two axes act distinctly. The PEO fraction controls the *strength* of pulse integration: the per-decade growth of the conductance ratio (the log–log slope α) falls with PEO content, most cleanly at the lowest salt, where it drops from $\alpha \approx 1.1$ (near-linear accumulation) at PEO 0.3 to $\alpha \approx 0.3$ (strongly compressive) at PEO 1.2; and the peak conductance ratio, i.e. the usable dynamic range, is largest in the low-PEO, low-salt corner ($\approx 480\times$ at PEO 0.3/salt 0.045) and falls to a few-fold at high PEO. The salt fraction, by contrast, controls *how far* the response can be driven before it saturates: at the lowest salt (0.045) every cell *turns over* (the conductance ratio peaks and then declines) by $N \sim 100\text{--}300$ pulses, whereas at the highest salt (0.18) the response grows monotonically through the largest counts tested ($N \approx 1000$), with

salt 0.09 intermediate. Two caveats are kept in view: the α trend is noisier at the higher salts, where the sparsely sampled PEO 0.3/salt 0.18 cell ($n = 2$, curation-recovered) departs from it; and the turnover sets a practical ceiling on the usable pulse count. Read constructively, these responses are a composition-tunable, predominantly compressive and (in the low-salt cells) non-monotone integration of the input pulse train, the nonlinear input encoding that ?? builds on.

Third, the fading-memory timescale varies with composition over roughly 2 s–20 s. The longest retention occurs in the low-PEO row (at PEO 0.3, $t_{1/2} \approx 19$ s at salt 0.09, with an identified $\tau \approx 25$ s); increasing the PEO fraction to 0.6 or 1.2 shortens the timescale to a few seconds. A single curation-recovered device at the still-lower PEO 0.15 level ($t_{1/2} \approx 22$ s) is consistent with this trend, though it lies outside the replicated grid and is not counted toward it. The independently computed instrument-side single-exponential time constant reproduces the same ordering. Within each cell, the fading-memory time constant varies by $\sigma(\ln \tau) \approx 0.85$ between nominally identical devices at fixed composition; film thickness, spin speed, anneal, and device age do not explain this spread (Appendix C; quantified in ??). ?? models the between-cell and within-cell distributions as a heterogeneous bank of fading-memory elements; the range across the whole device library is mapped in Appendix C. The same composition axis appears in potentiation strength: the growth exponent α falls monotonically with PEO fraction (cell medians $\approx 0.9 \rightarrow 0.7 \rightarrow 0.35$), with within-cell variation around each median.

Across the replicated grid, increasing the proportion of the ion-transport phase shortens the fading-memory time and is accompanied by smaller switching and potentiation responses, while salt content mainly controls when the response saturates. The PEO gradient is statistically established for the fading-memory time (per-device Spearman $\rho = -0.52$, permutation $p = 0.009$); the window and potentiation descriptors move in the same

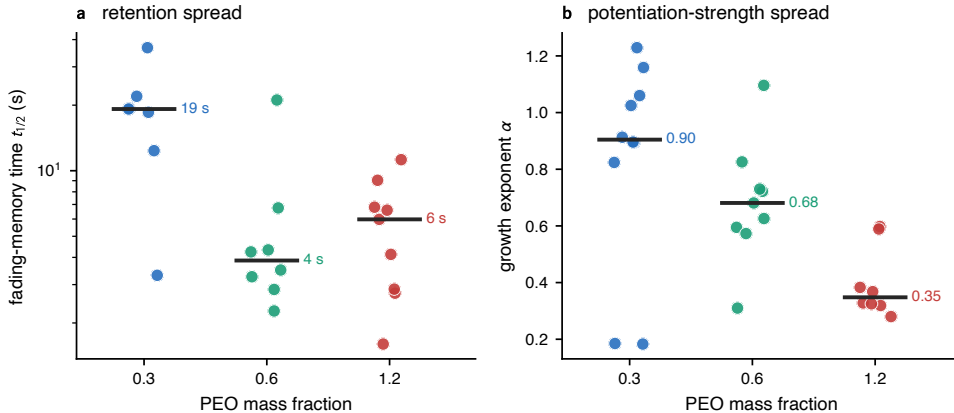


Figure 4.4: Per-device distributions underlying the cell medians of fig. 4.3 (*Super Yellow*/PEO/LiOTf/Ag; black bar is the cell median). (a) Fading-memory time $t_{1/2}$ and (b) potentiation growth exponent α , each against PEO fraction. The PEO fraction sets the broad level, while within-cell device-to-device variation is comparable to the steps between cells. ?? evaluates both sources of variation.

direction but remain marginal ($p \approx 0.06$ – 0.07). A two-way decomposition finds no significant interaction between PEO and salt, and bootstrap confidence intervals on every cell median are given in Appendix C. This replicated composition result is the quantitative core of the chapter; the per-cell values are tabulated in Appendix C. If a more ion-rich matrix relaxes a displaced ionic profile more readily, its equilibrium ionic resistance should also fall. Section 4.7 tests that prediction by impedance spectroscopy.

4.5 Chemical Tuning of the Dynamics

Beyond composition, the identity of the electrolyte components shifts the dynamics further. These comparisons are reported as an illustrative survey: each matched cell (fixed composition, electrode, and protocol) contains at most one or two screened devices, so the magnitudes below

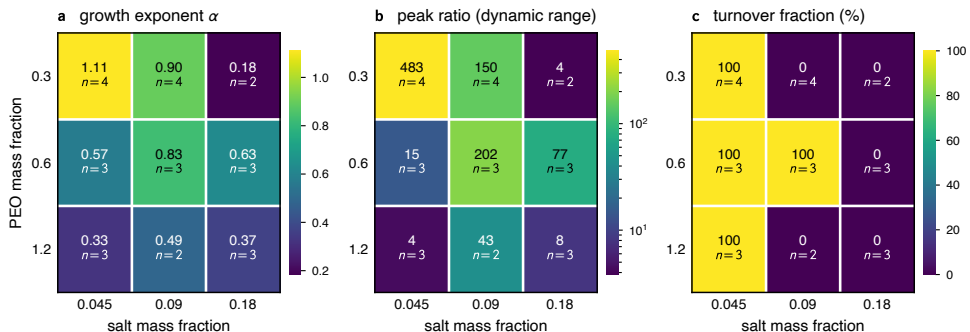


Figure 4.5: Potentiation (pulse-integration) response across the *Super Yellow*/PEO/LiOTf composition grid: (a) the log–log growth exponent α , (b) the peak conductance ratio (usable dynamic range, log scale), and (c) the fraction of devices whose response turns over within the measured range ($N \leq 1000$). Increasing the PEO fraction lowers α and the dynamic range; increasing the salt fraction suppresses turnover, sustaining growth to higher pulse counts. Per-cell median, with the device count n annotated in each cell.

are indicative and the sample size is given in every case. They are complemented here by structural and optical probes on *separate* sample sets (blend films and scratched powders at the lead composition, 0.3/0.09) examined by ATR-FTIR, X-ray diffraction, and UV-Vis absorption, which probe ion association, host backbone order, crystallinity, and the semiconductor’s electronic gap directly, independently of how a device is driven (figs. 4.6 and 4.8). These too are single specimens per chemistry and are read illustratively.

Host. Substituting the linear polyether PEO with the hyperbranched polyether TMPE, at fixed lithium-triflate chemistry and composition, shortens the fading-memory time by roughly a factor of six (PEO/LiOTf: $t_{1/2} \approx 20$ s–25 s, three screened silver devices; TMPE/LiOTf: ≈ 4 s, one screened silver device with an identified decay). This comparison has three PEO devices but only one TMPE device. The lithium pair from the designed host $\times$ cation batch could not be used because the PEO/LiOTf

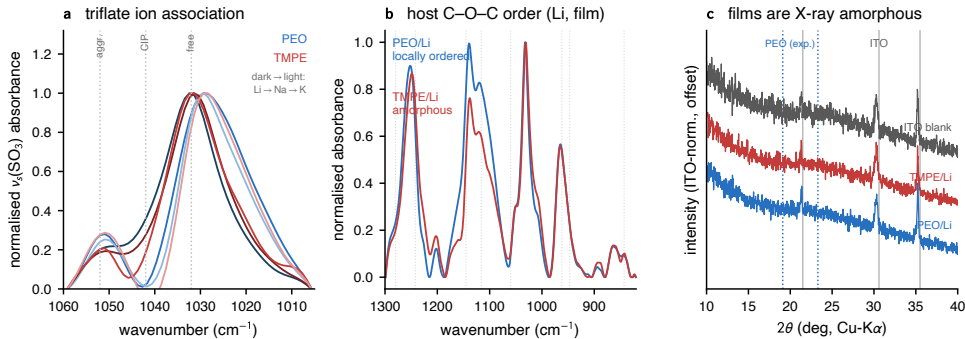


Figure 4.6: Microscopic corroboration of the chemistry axis (electrode-free *Super Yellow*-blend films/powders, 0.3/0.09, one specimen per chemistry). (a) ATR-FTIR triflate ion-association band $\nu_s(\text{SO}_3)$, whose position reports the ion-association state (spectroscopically free near 1032 cm^{-1} , contact ion pair near 1042 cm^{-1} , aggregate above; reference positions dotted [2]; PEO hosts in blues, TMPE hosts in reds, darker to lighter for Li/Na/K): it lies at 1029 cm^{-1} – 1032 cm^{-1} in every host $\times$ cation sample, with no cation-resolved shift. (b) ATR-FTIR C–O–C envelope: the linear PEO host (blue) shows a resolved multiplet, indicating retained *local* (conformational) ether-backbone order, whereas the hyperbranched TMPE host (red) shows a single broad, amorphous band [2]. (c) X-ray diffraction of PEO/Li and TMPE/Li with a bare-ITO blank: all patterns show the crystalline ITO substrate reflections (grey lines; 21.5° , 30.6° and 35.5°), with no sharp PEO crystalline doublet (blue dotted; 19.1° and 23.3°) or crystalline salt phase. The composite films are therefore X-ray amorphous, and the PEO order in (b) is short-range [2, 3]. ATR spectra are normalised after a local/rubber-band baseline and XRD is normalised to the ITO peak; intensities are not comparable across specimens because sampling and thickness differ, so only band positions and shapes are interpreted.

device was broken. The analysis therefore compares the TMPE device with the nearest-matched silver PEO/LiOTf devices (same operator, fresh semiconductor, glovebox), which differ in fabrication date.

An independent microscopic correlate of the host substitution is visible in the polyether backbone itself, and at two length scales. In ATR-FTIR the linear PEO film shows a resolved C–O–C stretching multiplet, with the weak order-sensitive marker bands near 843 cm^{-1} , 962 cm^{-1} and

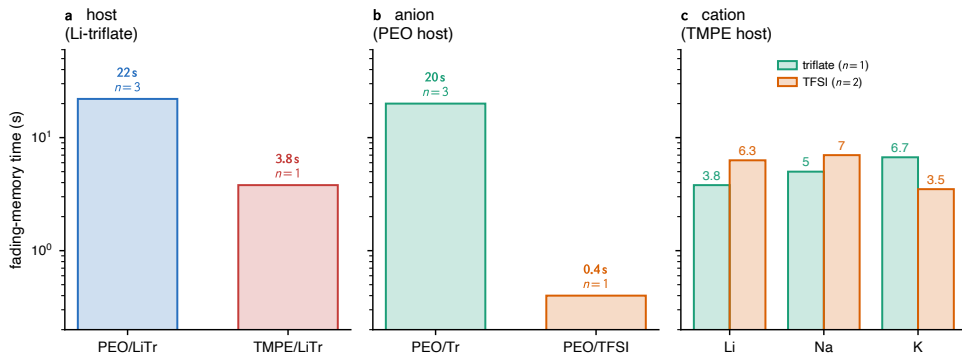


Figure 4.7: Illustrative effect of electrolyte chemistry on the fading-memory time constant (*Super Yellow*, 0.3/0.09, silver, matched 4 V/2 V protocol). Host (PEO vs TMPE, Li-triflate) and anion (triflate vs TFSI, PEO host) produce the largest shifts. The cation panel shows the same three cations in the TMPE host under each anion: potassium is longest in triflate and shortest in TFSI, so the cation produces no consistent, host- and anion-independent ordering. Sample sizes are small (n as labelled); see text.

1242 cm^{-1} , indicating retained *local* (conformational) order of the ether backbone, whereas the hyperbranched TMPE film collapses these into the single broad band of an amorphous polyether (fig. 4.6b) [2]. At the longer length scale of crystallites, however, X-ray diffraction on the same sample set finds *neither* host crystalline in the composite: every pattern is dominated by the ITO substrate (confirmed against a bare-substrate blank), with no sharp PEO crystalline reflections and no segregated crystalline salt phase (fig. 4.6c), so the films are X-ray amorphous and PEO retains short-range order without forming long-range crystallites [2, 3]. A host that is locally more ordered, even while globally amorphous, set against a fully amorphous TMPE, is consistent with the lower segmental mobility and slower ionic relaxation that the *longer* PEO fading-memory time would require: in polyether electrolytes ion transport is carried by segmental motion of the host above its glass transition, so the relaxation of a displaced ionic profile is rate-limited by that segmental mobility [4–6].

With one specimen per host this is offered as a structural correlate, not a demonstrated cause. The host ordering is moreover not specific to the silver electrode: on the independent gold-electrode corpus at the same composition, PEO again retains far longer than TMPE and supports a markedly wider switching window and stronger potentiation (section 4.8), so the host effect survives a change of electrode. That the salt forms no crystalline phase in any chemistry additionally supports reading these blends as homogeneous, salt-dissolved ion-conducting media.

Anion. Replacing triflate with the more charge-delocalised TFSI anion in the PEO host reduces the fading-memory time from ≈ 20 s to approximately 1 s. This direction is consistent with the electrolyte literature: charge delocalisation in TFSI weakens ion pairing and lowers salt lattice energy, promoting dissociation and plasticisation of the polyether host and increasing ionic mobility [7, 8]. The TFSI decays also tend toward a compressed shape, whereas the triflate decays are stretched, suggesting different relaxation-time distributions. This comparison is weakly controlled: the single PEO/TFSI device belongs to a later fabrication generation and also differs in spin speed and salt filtration. The approximately twenty-fold difference is therefore reported as illustrative.

Cation. The data do not support the hypothesised $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ ordering of fading-memory time derived from the hard-soft acid-base argument of ?? [9]. Two cation series use the same TMPE host, composition, electrode, and protocol, and differ in the salt anion (fig. 4.7). With triflate, the timescale increases from Li^+ (≈ 3.8 s) to Na^+ (≈ 5.0 s) and K^+ (≈ 6.7 s); each value comes from one screened device with $R^2 \geq 0.93$. With TFSI, potassium is shortest (≈ 3.5 s), while lithium and sodium are longer (≈ 6 s–7 s, two devices per cation). Thus, cation identity shifts the timescale by roughly a factor of two within a fixed anion, but the

ordering changes with the anion and within-cation variation is comparable to between-cation differences. These sample sizes cannot establish a host- and anion-independent ordering. A powered test would require a dedicated series at fixed host, anion, protocol, and electrode with at least three replicates per cell (section 4.10).

Independent probes provide additional constraints. The triflate symmetric SO_3 stretch $\nu_s(\text{SO}_3)$ lies at 1029 cm^{-1} – 1032 cm^{-1} in all six host $\times$ cation films and powders, a spread at or below instrumental resolution with no cation ordering (fig. 4.6a). UV-Vis absorption of *Super Yellow*/TMPE films places the *Super Yellow* π – π^* peak [10] at 446 nm for Li^+ , Na^+ , and K^+ (fig. 4.8). These results indicate no resolved cation effect on ion-pairing state or semiconductor optical gap; neither measurement tests the ordering of retention directly. On gold electrodes, the TMPE/triflate steady-state window is also flat across the three cations (on–off ratio ≈ 1.3 – 1.5 , no monotone trend; section 4.8).

In these illustrative comparisons, host and anion substitutions produce larger shifts in fading-memory time than cation substitution.

4.6 Protocol-Dependent Timescale

A methodological result constrains every comparison above. Because the read voltage is above the ionic activation threshold and write amplitude controls the displaced ionic profile, the *drive protocol* influences the apparent fading-memory time. In one device measured under both protocols, the fitted relaxation time increases from $\approx 4.6\text{ s}$ under a 3 V write and 1.5 V read to $\approx 15.5\text{ s}$ under a 6 V write and 3 V read (fig. 4.9).

One interpretation is that a larger write amplitude displaces more ionic charge and, at the largest amplitudes, may introduce electrode-level electrochemical processes, producing a longer-lived written state. The

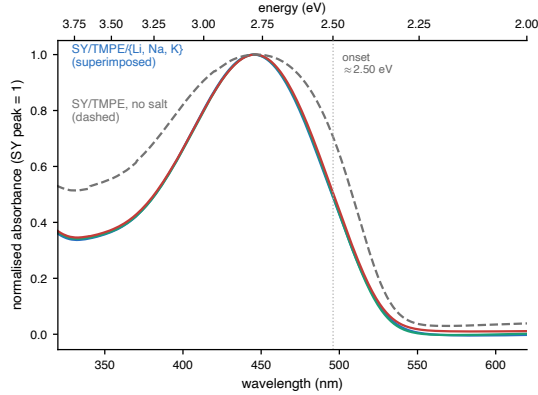


Figure 4.8: UV-Vis absorption of *Super Yellow*/TMPE blend films (lead composition 0.3/0.09; Au-electrode 2025 generation; one specimen per chemistry), normalised to the *Super Yellow* π - π^* peak. The peak (≈ 446 nm) and the absorption onset (≈ 2.50 eV) are superimposed for Li^+ , Na^+ , and K^+ ; the salt-free control (dashed) coincides at the band edge (its differing blue side and lower absolute absorbance are thickness/scattering, not electronic). The cation does not shift the semiconductor’s optical gap. Only band *positions* are read because absorbance magnitudes are thickness-confounded.

supra-threshold read may also perturb the state during interrogation. The composition result of section 4.4 remains internally comparable because it uses one fixed protocol. Chemistry comparisons spanning device generations are more exposed to protocol differences and are reported as illustrative. The apparent volatile timescale is therefore co-determined by material and drive conditions; cross-device or cross-chemistry comparisons must match the write/read protocol, electrode, and composition.

The same lever reappears, independently, on the gold-electrode corpus of section 4.8. There, matched-amplitude negative pulse trains sweep the pulse count and the inter-pulse interval together, with the relaxation recorded in the same run, and both halves of the protocol dependence are observed directly: potentiation grows steeply as the inter-pulse interval shortens (rank correlation $\rho \approx -0.94$ to -0.99 , $p < 0.001$, in three inde-

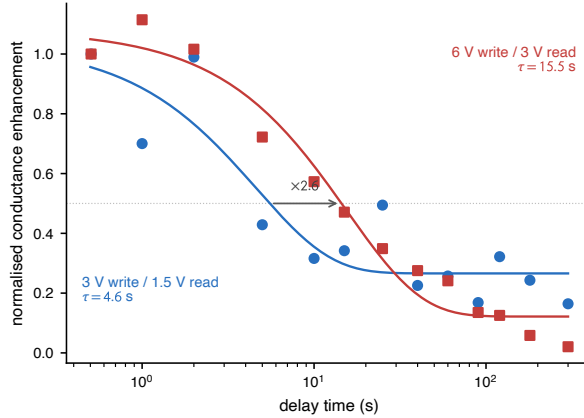


Figure 4.9: The apparent fading-memory time is set in part by the drive protocol. The same device, measured under a larger write/read amplitude, shows a markedly longer apparent relaxation time: the fitted single-exponential time constant rises from $\tau \approx 4.6$ s to ≈ 15.5 s, and the half-enhancement crossing (dotted, arrow) shifts by $\approx 2.8\times$.

pendent series of 9–11 matched runs), and the depth of the written state co-sets the subsequent relaxation time ($\rho \approx +0.46$ between potentiation ratio and decay half-time, $p = 0.015$, $n = 28$ runs). The drive–retention coupling that this section infers from a two-protocol contrast is therefore seen *within single experiments* on a second electrode (Appendix C).

4.7 Microscopic Corroboration: Impedance Spectroscopy

Every result so far rests on the *driven* response (pulses and decays). Small-signal impedance spectroscopy offers an independent test of the mechanism proposed for the composition trend: if increasing the PEO fraction shortens the fading-memory time because a more ion-rich matrix lets the displaced ionic profile relax more readily, then the *equilibrium* ionic resistance should fall along the same composition axis. The archive

contains such a measurement, acquired on the same devices: 0 V-DC spectra (40 mV AC, 1 Hz– 1×10^6 Hz) on 26 silver-electrode *Super Yellow*/PEO/LiOTf devices spanning the composition grid.

Each spectrum is summarised in two separate ways. The first is *model-free*: the real part of the impedance at the apex of the main Nyquist semicircle, $Z_{\text{real}}^{\text{apex}}$, which fixes the resistance scale of the spectrum without committing to any circuit. The second is *model-based*: the bulk ionic resistance R_{ion} of an equivalent circuit adapted from the polymer light-emitting electrochemical cell analysis of Munar *et al.* [11], reported only as corroboration of the model-free quantity because the per-spectrum circuit fits are genuinely ill-conditioned. The circuit topology, its two automated fits, and the per-cell values of both observables are detailed in Appendix C.

Both observables fall steeply and monotonically with PEO fraction (fig. 4.10a). The model-free $Z_{\text{real}}^{\text{apex}}$ drops by more than three orders of magnitude across the grid (cell medians from $\sim 10^8 \Omega$ at PEO 0.15–0.3 to $\sim 10^5 \Omega$ at PEO 1.2), with a rank correlation against PEO of -0.69 per device ($n = 26$) and -0.83 on cell medians; the fitted R_{ion} gives the same picture (-0.84 on cell medians). This is the direction the mechanism requires: more ion-transport polymer, lower ionic resistance.

The EIS resistance scale and the *independently* measured fading-memory time fall along the *same* composition axis (fig. 4.10b): both decline together as the PEO fraction rises, and across cells $\log Z_{\text{real}}^{\text{apex}}$ covaries with $\log t_{1/2}$ at $r \approx +0.74$ ($+0.61$ per device, $n = 23$). Three qualifications bound this reading:

- The PEO fraction is the *common cause* of both quantities, so this correlation confirms that the same composition axis moves the equilibrium ionic resistance and the driven retention together; it is not evidence of a second, independent factor.

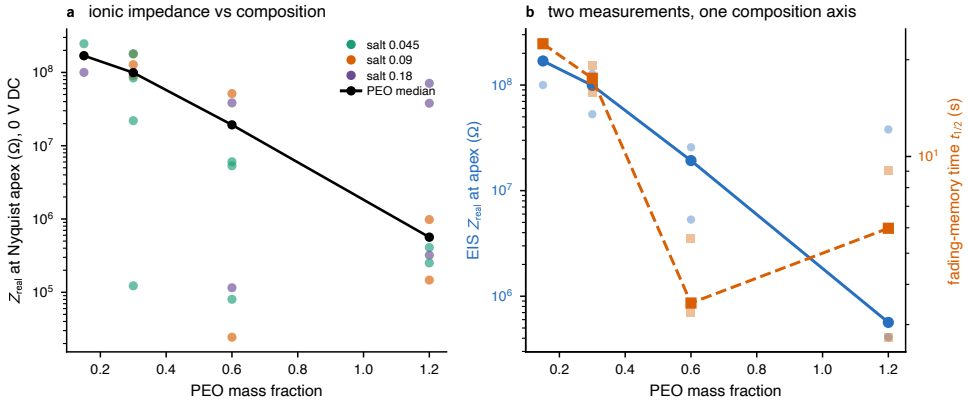


Figure 4.10: Equilibrium (0 V-DC) impedance spectroscopy corroborates the composition control of the dynamics (silver *Super Yellow*/PEO/LiOTf grid). (a) The impedance resistance scale (here the model-free real impedance at the Nyquist apex) falls by more than three orders of magnitude as the PEO fraction rises (points are devices, coloured by salt; the black line is the PEO median). (b) The equilibrium impedance scale (left axis) and the *independently* measured fading-memory time $t_{1/2}$ (right axis) decline along the *same* PEO composition axis (cell-level points with per-PEO median trends): the two are co-driven by composition, not independent variables.

- The resistance *scale* carries the composition signal, whereas the EIS relaxation *frequency* does not: the apex frequency corresponds to a sub-second RC time and does *not* track the seconds-long fading-memory time. This separates the small-signal ionic-transport bottleneck reported by impedance spectroscopy from the slower large-signal relaxation that determines the fading-memory response. The two timescales are nonetheless reconcilable in *magnitude*: the sub-second apex RC is the geometric-capacitance response, and scaling it by the chemical-to-geometric capacitance ratio measured independently from the low-frequency branch ($\gtrsim 33$) predicts a large-signal relaxation of several seconds, inside the observed $t_{1/2}$ range (Appendix C).

- The effect is not a film-thickness artefact: $Z_{\text{real}}^{\text{apex}}$ spans $\sim 10^4\times$ while the film thickness spans only $2.6\times$, and the PEO dependence survives controlling for thickness (partial correlation -0.60 ; section 4.2).

Impedance spectroscopy thus supplies, for the ionic-resistance half of the mechanistic picture below, the independent microscopic support that the driven measurements alone cannot.

4.8 Electrode and Polarity Dependence

Every result so far uses the silver top electrode. The archive also contains 50 gold-electrode devices, treated here as a third illustrative axis. Every gold device uses the single lead composition (PEO or TMPE host, 0.3/0.09); this corpus therefore contains no information about the composition series of section 4.4, which remains a silver-only result. It instead contrasts electrochemically active silver with inert gold at matched chemistry and provides an independent-electrode check on the chemistry comparisons of section 4.5.

At matched chemistry and composition, and under the canonical positive-polarity protocols used everywhere above, the electrode shifts all three behavioural families, and does so coherently (fig. 4.11). In the lead PEO/LiOTf cell, replacing silver by gold narrows the steady-state switching window (median on-off ratio $3.3 \rightarrow 2.0$, normalised loop area $0.33 \rightarrow 0.12$), weakens and flattens potentiation (peak conductance ratio $\approx 124\times \rightarrow 29\times$, log-log slope $\alpha \approx 0.9 \rightarrow 0.45$), and raises the activation voltage ($\approx 2.4\text{ V} - 2.7\text{ V}$), yet *lengthens* the fading-memory time, from $t_{1/2} \approx 19\text{ s}$ on silver (three devices) to $\approx 87\text{ s}$ on the two clean gold devices ($R^2 \approx 0.97 - 0.99$, $\beta \approx 0.6$). The same window-and-potentiation contraction holds in the TMPE/LiOTf cell (on-off $3.8 \rightarrow 1.3$; peak $54\times \rightarrow 1.9\times$),

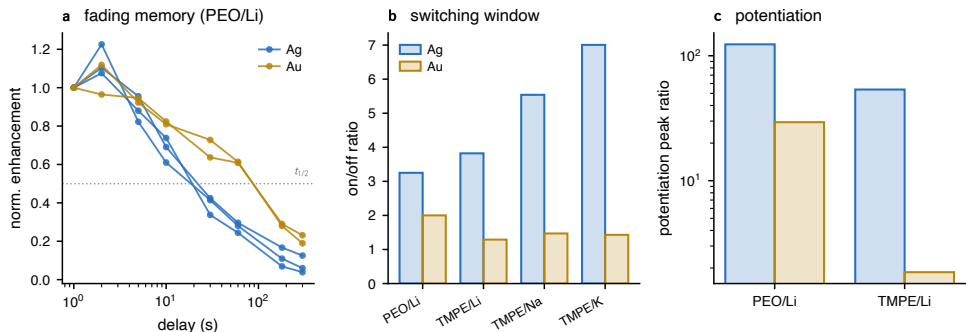


Figure 4.11: Illustrative electrode contrast at the lead composition (0.3/0.09), under the canonical positive-polarity protocols (for the negative-polarity behaviour of the gold devices see fig. 4.12). (a) Fading-memory decays of PEO/LiOTf devices, normalised to the first read: the gold curves (gold) cross half-enhancement later than the silver ones (blue), i.e. gold retains *longer*. (b) The steady-state switching window (on-off ratio) is narrower on gold for every chemistry, and is flat across the TMPE cation series on gold (no Li/Na/K ordering). (c) Pulse-potential dynamic range (peak ratio, log scale) is far smaller on gold. The inert electrode thus gives a weaker, higher-threshold, but more retentive response. Small n per cell, no human curve curation on the gold side, and a generation offset between the corpora bound the comparison (see text).

again at positive drive. The per-cell metrics for every chemistry on both electrodes, including the host-free and salt-free negative controls, are tabulated in Appendix C.

This difference is consistent with the contrast between active and inert electrodes. Silver can inject and electrodeposit mobile Ag^+ , the basis of electrochemical-metallisation switching, and may therefore add a large volatile component: a wider window, steeper potentiation, lower threshold, and faster relaxation [12, 13]. Gold is noble and inert, leaving the polymer-electrolyte ionic mechanism to produce a smaller, higher-threshold, and more persistent response. The higher activation voltage on thinner gold devices supports an interfacial electrode contribution over a film-thickness explanation [14].

Negative-polarity response of the gold devices. Additional campaigns on the same Au/TMPE/triflate generation measured negative bias. These data were recovered from outside the curated archive and analysed with a dedicated parser; their sources, validation against catalogued sweeps, and per-file tables are documented in Appendix C. The devices are strongly polarity-asymmetric: within a junction, bipolar sweeps carry $\approx 2.5\text{--}42\times$ more current at -3 V than at $+3\text{ V}$ (fig. 4.12a), with the same direction across the catalogued negative-going sweeps. Under negative drive, the gold devices show a rate-dependent switching window. Across sweep rates of $\sim 0.003\text{--}0.4\text{ V s}^{-1}$, the on-off ratio rises as the sweep slows (rank correlations down to -0.99 , $p < 0.001$), from ≈ 1 at the standard rate to $20\text{--}31$ in the slowest -3 V sweeps (fig. 4.12b). Constant-bias holds locate an activation threshold between -1.5 and -2.5 V : current decays below this range and grows by one to three orders of magnitude above it over tens to hundreds of seconds. Negative pulse trains potentiate by one to two orders of magnitude at a matched protocol (-5 V pulses, -2.25 V read: cation medians $\approx 40\text{--}230\times$), while positive pulses depress the written state. The response is therefore bidirectionally programmable under this negative-polarity protocol.

These campaigns add fading-memory measurements for the TMPE host on gold. Sparse delayed reads after the pulse trains, with the device floating between reads, follow strongly stretched exponentials ($\beta \approx 0.33\text{--}0.5$) with model-free half-decay times of $\approx 0.4\text{ s--}21\text{ s}$ across the three cations (fig. 4.12c). These values are on the same seconds scale as TMPE on silver and below the $\approx 87\text{ s}$ measured for PEO on gold, providing a gold-electrode comparison of host retention. Under the matched negative protocol, all three cations have overlapping potentiation ranges and relaxation times, with substrate-to-substrate spread comparable to any cation difference. The concentration of the response in one bias direction after removal of the active Ag^+ pathway is consistent with an interfacial role for the

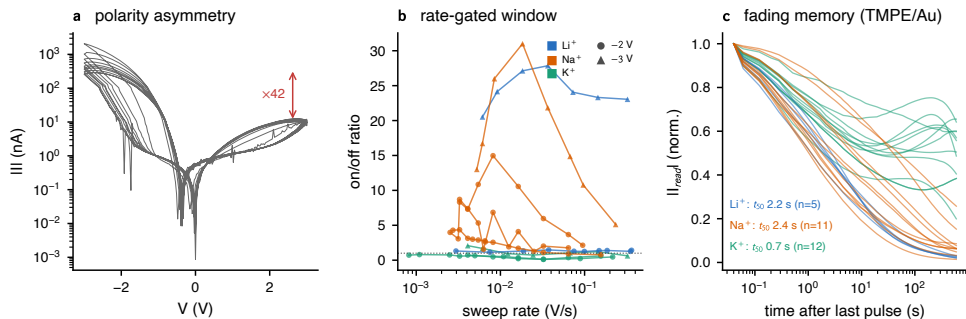


Figure 4.12: Negative-polarity response of the gold/TMPE/triflate devices (illustrative tier; Appendix C). (a) Exemplar bipolar sweep (± 3 V, ten cycles, Na^+ device): the negative branch carries $\approx 42\times$ the current of the positive one. (b) On–off ratio against sweep rate for all step-count series (colour = cation, marker = amplitude): slow sweeps give ratios of 20–31, while the standard fast protocol gives ≈ 1 . (c) Sparse-read relaxation after -5 V pulse trains, normalised to the first read, with per-cation median half-decay times. The times are on the seconds scale, as for TMPE on silver, and below those of PEO on gold.

dissimilar ITO and gold contacts and their accumulated ion profiles, but the electrical data do not establish that mechanism.

The second electrode reproduces two observations from section 4.5: PEO retains longer than TMPE, and the switching window has no transferable cation ordering. The gold corpus also provides negative controls absent from the silver set. Host-free (semiconductor-plus-salt) and salt-free (semiconductor-plus-host) films show neither a switching window (on–off ≈ 1.0) nor potentiation (peak ≈ 1), indicating that both the ion-transport polymer and dissolved salt are required for the memristive response.

Frequency-domain response: a bias-programmed low-frequency amplifier. A third corpus measures the same Au/TMPE/triflate chemistry in the frequency domain: 549 lock-in sweeps from 1 Hz to 500 kHz on a fresh batch, with one substrate per cation tracked from day 1 to day 23. The junction is placed in series with a 4 nF capacitor and driven by a small

sinusoid superimposed on a programmable DC offset. The apparatus, channel decode, and per-file tables are documented in Appendix C. Above ~ 10 Hz, the unbiased junction behaves as a passive linear dielectric: the transfer ratio follows the capacitive divider and is amplitude-independent to within a median 0.2 % from 0.5 to 2.4 V_{pk}. This boundary is consistent with the rate dependence in fig. 4.12b. In the same plateau, junction capacitance increases $\text{Li} < \text{Na} < \text{K}$ by $\approx 1.4\times$ across the same-batch trio, compared with an $\approx 1\%$ pixel spread. Same-recipe profilometry makes a thickness origin less likely, but one substrate per salt limits this result to a candidate dielectric cation effect; profilometry on matched specimens would test it directly. Under DC bias, the fundamental-frequency transfer below ~ 10 Hz increases from $|H| \approx 0.5$ without bias to ≈ 18 at ± 2.5 V (fig. 4.13). The bias-modulated ionic conductance transfers energy from the DC offset into the signal band. This gain occurs for all three cations at every age tested, is confined to the sub-10 Hz band, and is nearly symmetric in bias polarity (Appendix C). A biased junction therefore operates as a voltage-programmable low-pass amplifier over a band that overlaps the physiological signals analysed in ??.

All three gold corpora share the same illustrative-tier bounds, which the SI states for each corpus (Appendix C). Each uses the single lead composition on one or two substrates per cation; the curves are uncured; and the supra-threshold negative read makes decay times protocol-specific, allowing only order-of-magnitude comparison with silver. The 2024–2025 gold generation is also partly confounded with electrode because the silver chemistry was fabricated earlier and the co-fabricated silver batch was not measured under the decay protocol. Electrode and polarity are therefore treated as illustrative effects and cross-electrode checks on the chemistry axis. A matched-generation batch split between silver and gold is required for a quantitative electrode comparison.

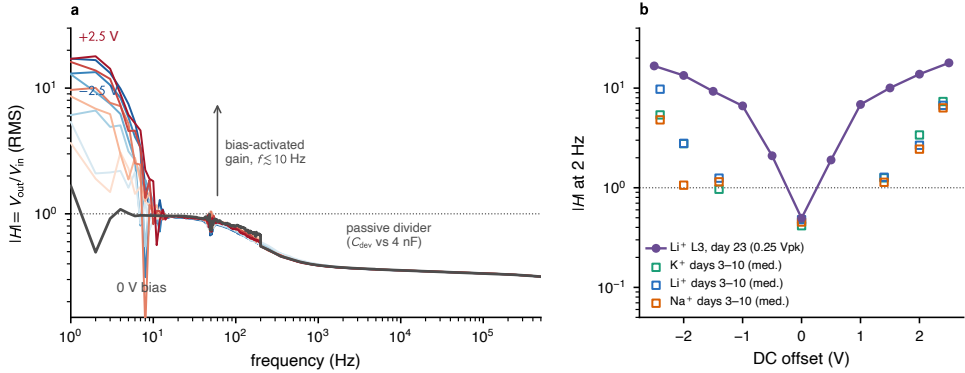


Figure 4.13: Frequency-domain response of the gold/TMPE/triflate devices (illustrative tier; Appendix C). (a) Transfer ratio $|H|(f)$ of one Li^+ junction in series with 4 nF (day 23, 0.25 Vpk) at eleven DC offsets from 0 (grey) to ± 2.5 V (blue negative, red positive). Above ~ 1 kHz, all curves follow the passive capacitive divider; below ~ 10 Hz, bias produces a fundamental-frequency gain of up to ≈ 18 , increasing with bias magnitude and nearly symmetric in sign. (b) $|H|$ at 2 Hz against DC offset for the day-23 series (line) and per-cation medians of earlier sessions (squares; days 3–10, 0.4 Vpk–1.4 Vpk). Gain increases with bias and occurs for all three cations.

4.9 Discussion

Composition is the only replicated axis for control of the volatile dynamics. Increasing the ion-transport-polymer fraction shortens the fading-memory time and is accompanied by smaller switching and potentiation responses. Host and anion substitutions shift the timescale over a wider range, but those comparisons are illustrative. Cation identity also shifts the dynamics, although the direction changes with host, anion, and protocol at the available sample sizes; the predicted $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ order is therefore not transferable across the dataset.

The electrode acts jointly with drive polarity. Relative to active silver, inert gold weakens switching and potentiation and lengthens retention under the canonical positive drive. Under negative drive, the same gold

devices show a full, rate-gated, bidirectionally programmable response (section 4.8). The frequency measurements locate the corresponding spectral boundary: above ~ 10 Hz the gold junctions behave as passive linear capacitors at every tested amplitude, whereas a DC bias produces programmable low-frequency gain below that boundary.

The hard–soft acid–base model of ?? remains a qualitative interpretation. It anticipates the cation ordering in the TFSI family, where the most weakly coordinating cation relaxes fastest, but the triflate family runs in the opposite direction. The present data therefore do not support a quantitative cation law.

Mechanistic interpretation. The mechanism remains a hypothesis. Increasing the PEO fraction does not produce the decrease in absolute on-state conductance expected from simple dilution of the *Super Yellow* percolation pathway; instead, on-state conductance rises by more than three orders of magnitude across the grid (Appendix C). A rising conductance floor is consistent with the contracting window: the ion-rich phase may add a parallel, weakly switchable path, with off-state conductance rising faster than on-state conductance. Faster relaxation of the displaced ionic profile could account for the accompanying decrease in fading-memory time. Salt content may set the number of mobile, chargeable states available before saturation, consistent with the later turnover at higher salt fraction.

Host and anion effects may arise from cation–ether coordination, ion pairing, and segmental mobility [5, 8], which set the distribution of local activation barriers behind stretched-exponential relaxation [1]. Equilibrium impedance supports the ionic-resistance part of this interpretation: it falls along the composition axis and tracks the fading-memory time (section 4.7). ATR associates the host effect with local ether-backbone order, retained in PEO and absent in amorphous TMPE, while XRD

shows that every composite remains X-ray amorphous and that the salt is dissolved rather than segregated. Across the cation series, ATR shows no corresponding ion-association fingerprint and UV-Vis shows no shift of the *Super Yellow* optical gap (section 4.5). These measurements are consistent with electrolyte-controlled ion transport rather than a change in the *Super Yellow* electronic backbone.

The anion and barrier-distribution links have not been established microscopically. Testing them requires temperature-dependent relaxation measurements and chemistry-resolved impedance (section 4.10); the hard-soft acid-base interpretation therefore remains qualitative.

The measured spread of fading-memory times can populate a physical reservoir with distinct temporal responses [15–17]. Across compositions and nominally identical devices, the timescale spans roughly an order of magnitude. A two- σ criterion on the cell-median span of $\Delta \ln t_{1/2} \approx 2.1$, relative to within-cell scatter, resolves approximately two to five composition bands (Appendix C). Each junction reproduces its switching window to $\approx 15\%$ sweep-to-sweep (Appendix C), substantially less than the diversity across the bank. Figure 4.14 places every screened silver device in a fading-memory ($t_{1/2}$) versus dynamic-range (peak ratio) plane. The PEO fraction sets the broad region occupied by each device, so one composition selects a paired memory/nonlinearity operating point; the remaining within-composition scatter supplies additional device-to-device diversity. ?? evaluates both sources of variation.

4.10 Limitations and Future Work

The principal limitation is statistical power outside the composition axis. Only the *Super Yellow*/PEO/LiOTf composition grid is replicated; the host, anion, and cation comparisons rest on one or two screened devices

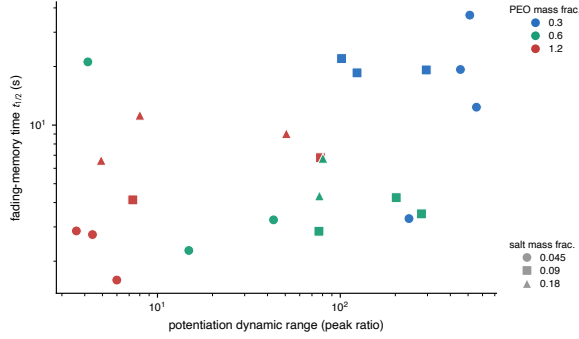


Figure 4.14: Per-device design space of the *Super Yellow*/PEO/LiOTf/Ag composition grid: fading-memory time $t_{1/2}$ against potentiation dynamic range (peak ratio), coloured by PEO fraction and marked by salt fraction (one point per device, median over its pixels). The PEO fraction selects the broad region a device occupies: low PEO (blue) in the long-memory, high-dynamic-range corner, high PEO (red) in the opposite one, so memory and nonlinearity are *paired*, not independently tunable. The substantial within-composition scatter is the genuine device-to-device heterogeneity that ?? exploits.

per matched cell and remain illustrative. Five consequences follow:

1. The cation hypothesis is not tested here at adequate power. The cation does shift the dynamics, but not as predicted: the triflate retention series runs $K > Na > Li$, the reverse of the hypothesised $Li^+ > Na^+ > K^+$, and re-orders again with the anion, so the hypothesised ordering is not what the data show and no transferable cation law can be fixed at one or two devices per cell.
2. The supra-threshold read and the amplitude dependence of the apparent timescale (section 4.6) mean that absolute retention values are protocol-specific.
3. The electrode is reported as an illustrative axis (section 4.8), but it remains partly confounded with fabrication generation, and device ageing is an uncontrolled variable that was not modelled.

4. Pulse integration and state relaxation were characterised *separately and at different write amplitudes*: the potentiation curves used a 3 V write and the fading-memory decays a 4 V write (table 4.1), at an effectively common inter-pulse interval. On the lead silver composition, the archive contains no experiment that varies input timing to probe their interaction directly. A matched-amplitude experiment exists on the gold/TMPE corpus (section 4.8), but it uses a different host, electrode, polarity, and read, so it establishes the direction of coupling rather than the lead-cell parameters.
5. The fading-memory protocol measured a single junction per device, so repeat-decay reproducibility of $t_{1/2}$ is not characterised at the population level (only one device carries two junctions; Appendix C). Within-device reproducibility is established only for the steady-state switching window, from its repeated sweeps.

Because drive amplitude affects the apparent timescale (section 4.6), combining the 3 V potentiation shape with the 4 V decay in ?? requires an explicit modelling assumption that the regimes are compatible. A matched-amplitude experiment that varies both pulse count and inter-pulse interval would test this assumption. Its gold/TMPE negative-polarity counterpart shows coupling between the two measurements (section 4.6); the lead-composition silver experiment remains future work.

A dedicated cation series in one host and anion, at a fixed protocol and electrode, with a sub-threshold read voltage (as in ??) and at least three replicates per cell, would turn the illustrative chemistry survey into a powered comparison. The impedance study of section 4.7 could also be extended beyond the replicated PEO/LiOTf axis at 0 V DC. The archive contains unmodelled bias-resolved spectra up to 3 V DC and chemistry-varied devices; analysing them would test the proposed host, anion, and barrier-distribution effects. A matched-generation batch split

between silver and gold top electrodes would isolate the electrode effect from fabrication generation.

Temperature-dependent relaxation measurements are needed to test the proposed role of segmental mobility and distributed activation barriers. A $\tau(T)$ series yielding an Arrhenius or Vogel–Fulcher–Tammann activation energy would connect the host and anion differences to a segmental-relaxation barrier and test the temperature dependence of the stretch exponent β . Encapsulation and systematic ageing measurements would address the remaining stability questions.

4.11 Inputs to Temporal Computing

In the replicated *Super Yellow*/PEO/LiOTf grid, composition tunes the fading-memory time over roughly 2s–20s, with the switching window and potentiation strength moving in the same direction. Illustrative host and anion contrasts extend the measured range. Cation identity also shifts the dynamics, but the available data do not establish the predicted $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ order as a transferable law. Drive protocol co-sets the apparent timescale and must remain fixed in quantitative comparisons.

?? uses the fading-memory times and device-to-device variability measured in this chapter. It also uses the pulse-update curves, including their saturation and turnover. Because the PEO fraction changes both the fading-memory time and the input nonlinearity α (fig. 4.14), each composition defines a paired operating point rather than two independent settings. Salt fraction provides a partly separate control of dynamic range and turnover. The simulations therefore compare banks drawn from one composition with banks spanning several compositions.

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